



STATISTICAL INFERENCE AND LEARNING

Academic Year: 2025/2026

Statistical Inference and Learning

Project

Car Insurance

10th June 2026

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1 Dataset and Data Preparation

This project analyzes a car insurance dataset containing 10,000 policyholder records. The objective is to identify factors associated with the probability of filing a claim, modeled through the binary response variable `OUTCOME` (1 if a claim was filed, 0 otherwise). Explanatory variables include demographic (`AGE`, `GENDER`, `RACE`, `EDUCATION`, `INCOME`, `MARRIED`, `CHILDREN`, `POSTAL_CODE`, `CREDIT_SCORE`), driving history (`DRIVING_EXPERIENCE`, `SPEEDING_VIOLATIONS`, `DUIS`, `PAST_ACCIDENTS`, `ANNUAL_MILEAGE`), and vehicle details (`VEHICLE_OWNERSHIP`, `VEHICLE_YEAR`, `VEHICLE_TYPE`). During preparation, the identifier variable was removed, and categorical predictors (including `POSTAL_CODE`) were converted into factors. Observations with missing values in `CREDIT_SCORE` and `ANNUAL_MILEAGE` were excluded, resulting in a final clean dataset of 8,149 observations. The response variable is moderately balanced (31% claims, 69% no claims), making it suitable for logistic regression.

2 Exploratory Data Analysis

The exploratory analysis was used to identify the main variables associated with claim occurrence before fitting the statistical models.

Figure 1 shows that claim rates vary substantially across categorical predictors. Younger drivers and drivers with less experience show much higher claim frequencies. In particular, the claim rate decreases clearly as driving experience increases, suggesting that this variable may be one of the strongest predictors. Income and vehicle year also show visible differences: lower-income policyholders and owners of vehicles manufactured before 2015 present higher claim rates.

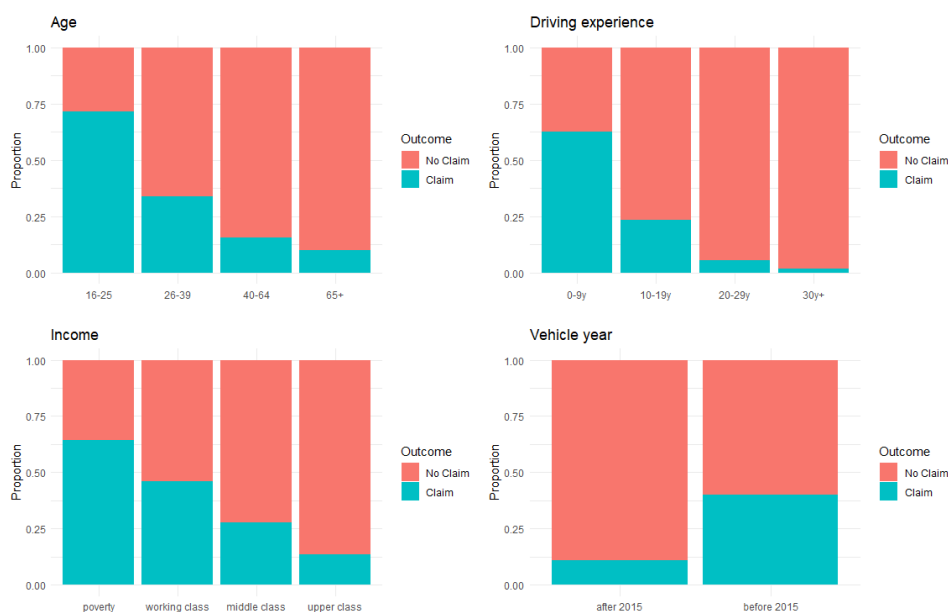
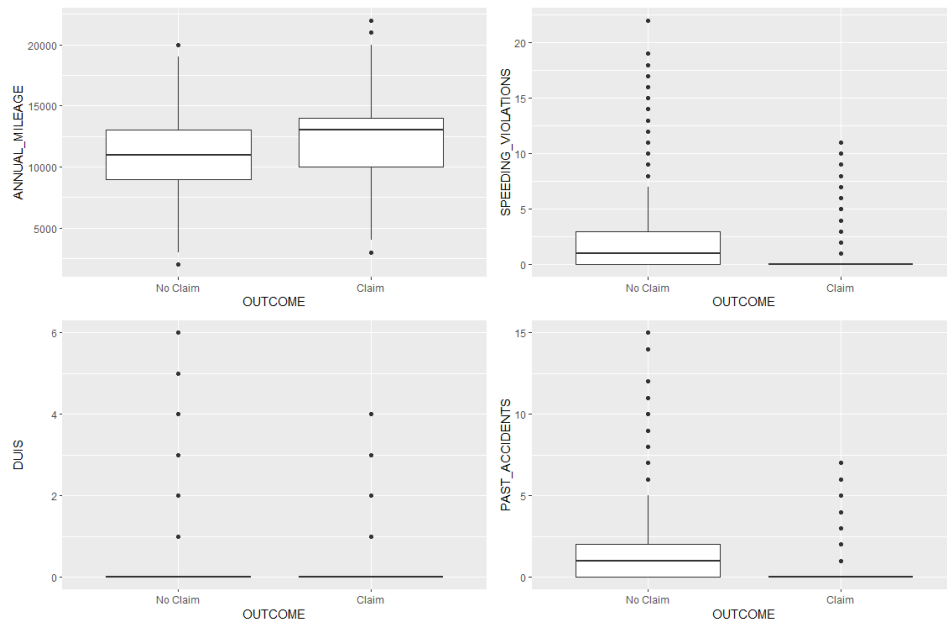


Figure 2 compares the numerical predictors by claim outcome. Policyholders who filed a claim tend to drive more miles per year, supporting the idea that higher road exposure increases claim probability.

For `SPEEDING_VIOLATIONS`, `DUIS` and `PAST_ACCIDENTS`, the interpretation is less direct. These variables are highly concentrated around zero, especially `DUIS`, where most observations have no recorded events. The boxplots show some higher values mainly among policyholders without claims, but this pattern is driven by a limited number of observations and should not be interpreted as evidence that violations or past accidents reduce claim risk.

A possible explanation is confounding with driving experience: more experienced drivers have had more time to accumulate past violations or accidents, but they may still have a lower current probability of filing a claim. Therefore, these exploratory results should be considered only descriptive and must be checked through multivariable models.



Overall, the EDA suggests that age, driving experience, income, vehicle year and annual mileage are associated with claim occurrence. For `SPEEDING_VIOLATIONS`, `DUIS` and `PAST_ACCIDENTS`, instead, the graphical evidence is weaker and should be interpreted with caution. These variables are mostly concentrated around zero, especially `DUIS`, and the apparent differences between claim and no-claim groups may be driven by a small number of observations.

Therefore, the exploratory results should not be interpreted as conclusive. They are useful to generate hypotheses, but multivariable models are needed to verify which predictors remain relevant after controlling for the others.

3 Logistic Regression and Model Selection

Since the response variable is binary, logistic regression was used to model the probability of filing an insurance claim.

The first model included only 'AGE' as predictor. This model showed a strong negative association between age and claim occurrence: older drivers had substantially lower odds of filing a claim compared with drivers aged 16–25. This result was consistent with the exploratory analysis.

A second model included both ‘AGE’ and ‘DRIVING_EXPERIENCE’. After adding driving experience, the coefficients associated with age became much smaller. This suggests that part of the apparent age effect observed in the EDA was actually explained by driving experience, since younger drivers are also those with less driving experience.

A third model was fitted including ‘AGE’, ‘DRIVING_EXPERIENCE’, ‘INCOME’, ‘CREDIT_SCORE’ and ‘VEHICLE_YEAR’. In this model, driving experience and vehicle year remained strong predictors, while credit score became non-significant. This shows the difference between marginal associations observed in the EDA and conditional effects estimated in a multivariable model.

Multicollinearity was assessed using the Variance Inflation Factor. All VIF values were low, suggesting that multicollinearity was not a major issue for the variables included in the model.

	GVIF	Df	GVIF ^{1/(2*Df)}
AGE	3.992962	3	1.259551
DRIVING_EXPERIENCE	2.727102	3	1.181998
INCOME	2.978935	3	1.199527
CREDIT_SCORE	2.223554	1	1.491159
VEHICLE_YEAR	1.092858	1	1.045398

After this preliminary analysis, the dataset was split into a training set and a test set. A full logistic regression model was fitted on the training set using all available predictors. Then, stepwise model selection based on AIC was applied to obtain a simpler model. The selected model retained driving experience, gender, vehicle ownership, vehicle year, marital status, postal code and annual mileage.

```
Call:
glm(formula = OUTCOME ~ AGE + DRIVING_EXPERIENCE + INCOME + CREDIT_SCORE +
    VEHICLE_YEAR, family = binomial, data = data_clean)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.55217    0.18811 -13.568  < 2e-16 ***
AGE.L         -0.19321    0.10259  -1.883   0.0597 .
AGE.Q         -0.04307    0.08050   0.535   0.5926 .
AGE.C         -0.02492    0.06992   0.356   0.7215
DRIVING_EXPERIENCE.L -2.84467    0.18819 -15.116  < 2e-16 ***
DRIVING_EXPERIENCE.Q  0.37153    0.15400   2.413   0.0158 *
DRIVING_EXPERIENCE.C  0.02846    0.10962   0.260   0.7951
INCOME.L      -0.64814    0.09734  -6.659 2.76e-11 ***
INCOME.Q      -0.02076    0.06263  -0.332   0.7402
INCOME.C       0.05808    0.06242   0.930   0.3521
CREDIT_SCORE  -0.37244    0.32982  -1.129   0.2588
VEHICLE_YEARbefore 2015 1.42633    0.08136 17.532  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 10105.7  on 8148  degrees of freedom
Residual deviance: 6829.7  on 8137  degrees of freedom
AIC: 6853.7
```

After the stepwise selection, the coefficients of the final logistic regression model were

transformed into Odds Ratios to improve interpretability. The table reports the estimated Odds Ratios and their confidence intervals.

Variable	OR	2.5%	97.5%
Intercept	0.106	0.061	0.184
Driving Experience (10–19y)	0.126	0.105	0.151
Driving Experience (20–29y)	0.015	0.011	0.022
Driving Experience (30y+)	0.008	0.003	0.015
Male	2.776	2.347	3.290
Vehicle Ownership (Yes)	0.139	0.117	0.166
Vehicle Year (before 2015)	7.852	6.296	9.856
Married (Yes)	0.753	0.622	0.911
Postal Code 21217	1.34×10^9	1.38×10^6	1.27×10^{40}
Postal Code 32765	3.990	3.241	4.926
Postal Code 92101	3.930	2.654	5.819
Annual Mileage	1.00014	1.00011	1.00018

Driving experience shows a strong protective effect. Compared with drivers with 0–9 years of experience, drivers with more experience have substantially lower odds of filing a claim. Male policyholders show higher odds of claim occurrence, while vehicle ownership is associated with lower odds. Vehicles manufactured before 2015 show much higher odds of claim occurrence compared with newer vehicles.

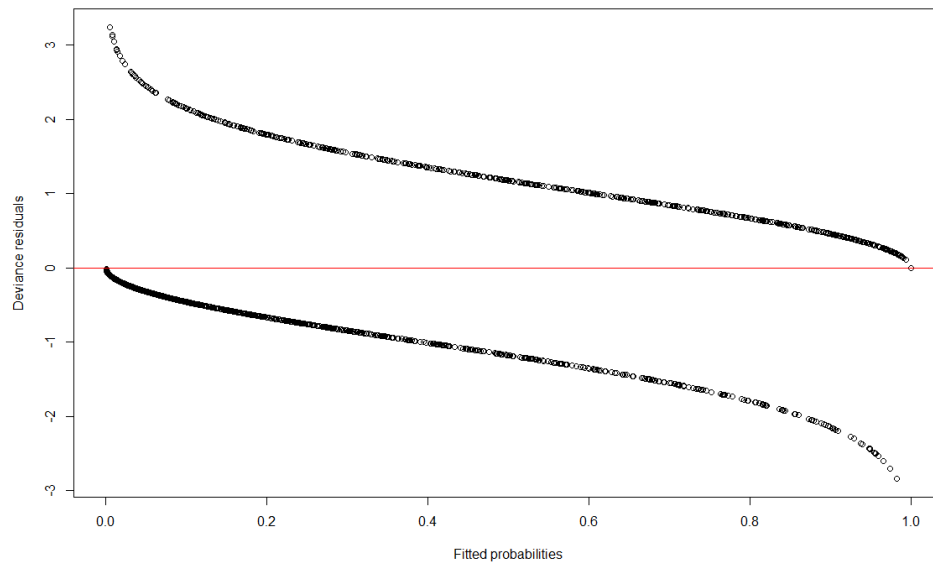
Postal code 21217 must be interpreted with caution. All observations in this category correspond to claims. For this reason, its coefficient and Odds Ratio are unstable and should not be interpreted literally.

After removing ‘POSTAL_CODE’, the main predictors remained statistically significant and consistent with the original model. The AUC decreased to 0.906, indicating that information improves prediction. However, the model still performed well, suggesting that the main conclusions are not driven only by ‘POSTAL_CODE’.

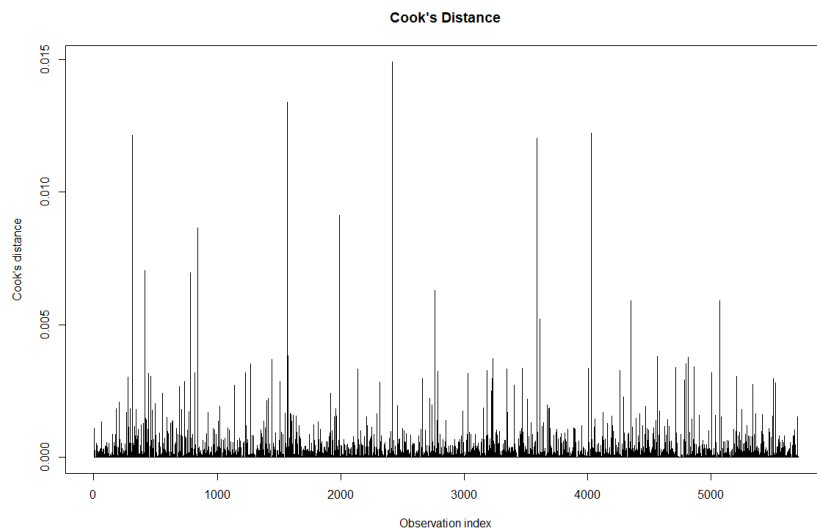
4 Model Diagnostic

After selecting the final logistic regression model, diagnostic checks were performed to assess the adequacy and stability of the model.

First, deviance residuals were plotted against the fitted probabilities. Since the response variable is binary, the residual plot shows the typical two-branch structure expected in logistic regression. No clear systematic pattern was observed, suggesting that the model does not show evident misspecification.



Cook's distance was then used to identify potentially influential observations. Although some observations exceeded the conventional threshold $4/n$, the overall Cook's distance values remained very small. Therefore, no single observation appeared to have an excessive influence on the fitted model.



Leverage values were also inspected to detect observations with unusual predictor combinations. Most leverage values were low, indicating that the model was not driven by a small number of extreme observations.

Finally, the model was evaluated on the test set using the ROC curve and the Area Under the Curve. This provides an out-of-sample measure of the model's ability to distinguish between policyholders who filed a claim and those who did not. The AUC was computed on the test set rather than on the training data. This is important because it evaluates the model on observations that were not used during model fitting, providing a more realistic estimate of predictive performance.

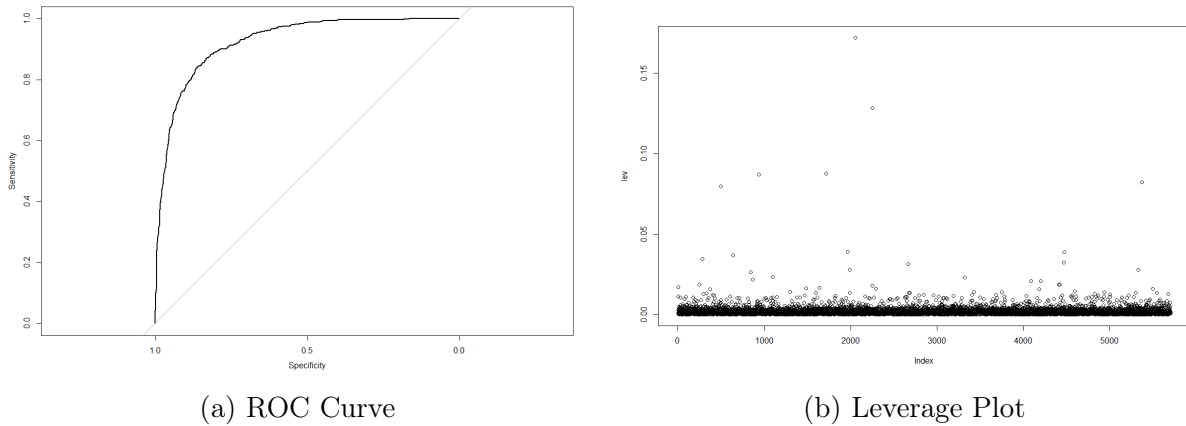


Figure 1: Model diagnostic plots

5 Robustness Check: Ridge and Lasso Regression

To assess the robustness of the variable selection process, Ridge and Lasso logistic regression models were fitted using the training set.

5.1 Ridge

Ridge logistic regression was fitted using all available predictors. The optimal value of λ was selected by cross-validation.

The Ridge model confirmed the main findings of the logistic regression analysis. The largest effects were associated with driving experience, vehicle ownership, vehicle year,

gender and postal code. Variables such as race, education, credit score and vehicle type had coefficients close to zero, suggesting a limited additional contribution.

Since Ridge does not perform variable selection, all predictors remain in the model, but less informative variables are strongly shrunk.

5.2 Lasso

Lasso is also fitted using the same training data. Unlike Ridge, Lasso can set some coefficients exactly equal to zero.

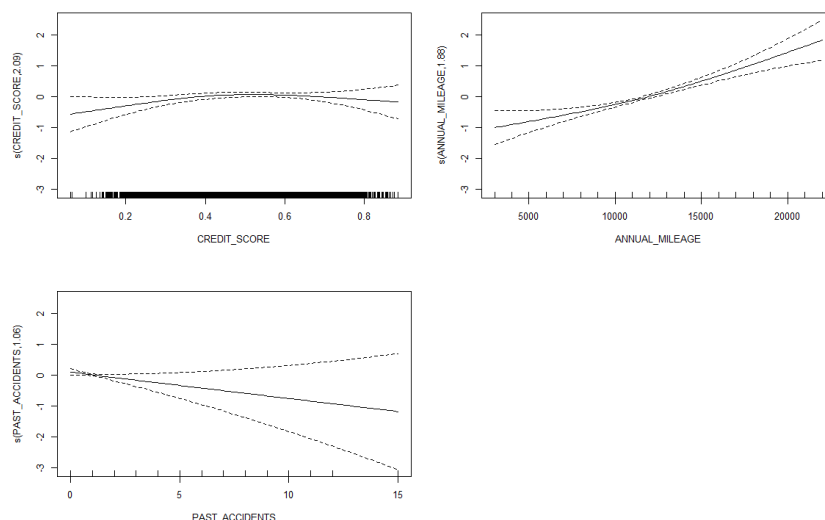
The Lasso model removed several weak predictors, including some levels of age, education, credit score, vehicle type and speeding violations. The variables retained by Lasso were largely consistent with those selected by the stepwise logistic regression model.

In particular, driving experience, vehicle ownership, vehicle year, gender, postal code and annual mileage remained important. This supports the robustness of the main conclusions obtained from the logistic regression analysis.

Overall, Ridge and Lasso did not substantially change the interpretation of the analysis. The most important predictors identified by the logistic regression model were also emphasized by the regularized models. This suggests that the final results are not driven only by the specific stepwise selection procedure.

6 Generalized Additive Model

A Generalized Additive Model (GAM) investigated nonlinear effects by applying smoothing splines to continuous variables (CREDIT_SCORE, ANNUAL_MILEAGE, PAST_ACCIDENTS) alongside the selected categorical predictors. Only ANNUAL_MILEAGE exhibited a significant nonlinear relationship, showing an increasing probability of claims as exposure on the road increases.



7 Final Model Comparison

Finally, the predictive performance of the different models was compared using the Area Under the ROC Curve (AUC). All models were evaluated on the same test set in order to obtain a fair comparison.

Model	AUC
Logistic Full	0.9243
Logistic StepAIC	0.9256
Ridge	0.9235
Lasso	0.9252
GAM	0.9255

The comparison shows that all models achieved high AUC values, indicating strong discriminative ability. More complex models such as Ridge, Lasso and GAM did not provide a substantial improvement over logistic regression.

For this reason, the stepwise logistic regression model was preferred as the final model. It provides a good balance between predictive performance and interpretability, while retaining a relatively small set of relevant predictors.

8 Conclusion

The analysis shows that claim occurrence is mainly associated with driving experience, vehicle characteristics, gender, annual mileage and postal code.

The EDA initially suggested that age, income and credit score were important predictors. However, after fitting logistic models, these effects became much weaker. In particular, the apparent effect of age was largely explained by driving experience.

The final stepwise logistic regression model achieved strong predictive performance on the test set and was preferred because it offered a good balance between accuracy and interpretability. Ridge, Lasso and GAM confirmed the main results, but did not provide a substantial improvement.

9 Extra: Classification

As an additional analysis, the problem was treated as a classification task. Logistic Regression, LDA, QDA, Naive Bayes and k-NN were compared on the same test set using the main predictors selected by the logistic model. For QDA, POSTAL_CODE was excluded because its dummy variables and sparse levels made the covariance estimation unstable.

Table 1: Classification model comparison

Model	Accuracy	Balanced Acc.	Sensitivity	Specificity
Logistic Regression	0.862	0.833	0.758	0.908
LDA	0.859	0.833	0.765	0.902
QDA	0.730	0.780	0.912	0.649
Naive Bayes	0.835	0.802	0.715	0.889
k-NN ($k = 37$)	0.867	0.846	0.791	0.901

Logistic Regression and LDA obtain almost identical results, while QDA performs worse because it predicts too many observations as *Claim*. k-NN achieves the best classification performance, but Logistic Regression remains preferable because it provides similar predictive quality with clearer interpretation through coefficients and Odds Ratios.

10 Future improvement

Future improvements could focus on a better treatment of missing values, using imputation instead of removing incomplete observations.

A second improvement concerns `POSTAL_CODE`. In this project, one postal code was associated only with claim observations, producing an unstable and extremely large estimate. Future work could handle this variable more carefully, for example by grouping rare postal codes, removing unstable categories.